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To cite this article: Seyyed Kazem Banihashem, Hojjat Dehghanzadeh, Douglas Clark, Omid Noroozi & Harm J. A. Biemans (2024) Learning analytics for online game-Based learning: a systematic literature review, Behaviour & Information Technology, 43:12, 2689-2716, DOI: [10.1080/0144929X.2023.2255301](https://doi.org/10.1080/0144929X.2023.2255301)

To link to this article: <https://doi.org/10.1080/0144929X.2023.2255301>



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Published online: 19 Sep 2023.



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Learning analytics for online game-Based learning: a systematic literature review

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ABSTRACT

Game-based learning researchers have been investigating various means to maximise learning in educational games. One promising venue in recent years has been the use of learning analytics in online game-based learning environments. However, little is known about how different elements of learning analytics (e.g. data types, techniques methods, and stakeholders) contribute to game-based learning practices within online learning environments. There is a need for a comprehensive review to bridge this gap. In this systematic review, we examined the related literature in five major international databases including Web of Science, Scopus, ERIC, IEEE, and compiled Proceedings of the International Conference on Learning Analytics and Knowledge. Twenty relevant publications were identified and analysed. The analysis was conducted using four core elements of learning analytics, namely the types of data that the system collects (what), the methods used for performing analytics (how), the reasons the system captures, analyzes, and reports data (why), and the recipients of the analytics (who). This study synthesises the existing literature, provides a conceptual framework as to how learning analytics can enhance online game-based learning practices in higher education, and sets the agenda for future research.

ARTICLE HISTORY

Received 29 December 2022
Accepted 1 August 2023

KEYWORDS

Conceptual framework;
higher education; game-
based learning; learning
analytics; systematic review

1. Introduction

Game-based learning (GBL) refers to a setting where learners develop knowledge and skills through game content and gameplay (Prensky 2003). Mayer (2020) states that GBL occurs when a change happens in one's skills or academic knowledge as a result of playing a game. Such knowledge and skill development are enhanced via game activities that create problem-solving spaces and challenges, providing learners/players with a sense of accomplishment (Qian and Clark 2016). Trends in educational research indicate that GBL has received a lot of attention in education (see All, Castellar, and Van Looy 2021; Ioannou 2021). Moreover, we have a plethora of evidence that GBL is effective in enhancing various cognitive and affective learning outcomes across educational settings including in higher education (Abdul Jabbar and Felicia 2015; Clark, Tanner-Smith, and Killingsworth 2016; Foster and Shah 2020; Huizenga et al. 2009; Lei et al. 2022; Tokac, Novak, and Thompson 2019). For instance, GBL has shown to be effective in improving learners'

learning performance (e.g. Hwang, Wu, and Chen 2012; Sung and Hwang 2013), argumentation skills (e.g. Noroozi, Dehghanzadeh, and Talaei 2020), language learning (e.g. Dehghanzadeh et al. 2021), and problem-solving skills (e.g. Eseryel et al. 2014; Hwang, Wu, and Chen 2012; Taub et al. 2020). Research has also shown positive impacts of GBL on learners' motivation and engagement (see Breien and Wasson 2021; Huang 2011; Huizenga et al. 2009; Partovi and Razavi 2019). This could be due to the entertaining nature of GBL, which entices learners into learning as they are playing and enjoying the game regardless of their underlying theory (Celestini 2020).

However, not all GBL environments are effective for learning. Well-designed GBL environments have clear goals, incremental or adaptive challenges, as well as ongoing feedback (Shute and Ke 2012). Well-designed GBL includes gaming activities such as rules and narratives to be followed, objectives to be achieved, challenges that need to be encountered, and information needed to overcome those challenges (Wang and Zheng 2021).

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Moreover, GBL environments may foster germane cognitive load (i.e. the type of cognitive load that helps learning), however, games can generate some extraneous cognitive loads that can distract learners from what is supposed to be learned (i.e. not related to the learning objectives of interest) (Mayer 2020). Well-designed educational games create a balance by including instructional and learning features that regulate the extraneous cognitive load and sufficiently generate the crucial germane cognitive load in learners (Mayer 2020; Shute and Ke 2012).

There are many reasons why well-designed GBL is effective: (a) educational games can highly motivate and engage learners. According to the Self Determination Theory (SDT, Deci and Ryan 2008), learners can have high levels of autonomy (through choices they make), mastery (through feeling competent as they level up), and relatedness (through their interactions with other learners) when playing digital games in general; (b) well-designed GBL environments (Shute and Ke 2012) facilitate *flow* (Csikszentmihalyi 1990), that is, the ultimate level of engagement in a task at hand one could lose track of time. Research shows that engagement in the learning task at hand is positively related to learning (e.g. Gresalfi and Barnes 2016); and (c) GBL is supported by many learning theories (e.g. behaviourism, cognitivism, and constructivism; Wu et al. 2012). For example, according to the situated learning theory (Lave and Wenger 1991), learners learn better when they learn the content or the skill of interest in a context that is near the situation they need to perform in real-life – GBL facilitates situated learning. Similarly, GBL is supported by other theories such as learning by doing (Anzai and Simon 1979) which indicates people learn better when they do something rather than know how to do something. Additionally, learners in behaviourist GBL contexts are expected to be aware of the objectives of the game and need a scenario to achieve these goals through stimuli-response processes. In a constructivist approach, GBL focuses on interactions where the learners stand at the heart of the game and construct knowledge as they engage with the game via interactions with the content and other learners (Wu et al. 2012).

Given the evidence regarding the effectiveness of GBL, researchers have been trying to maximise learning in educational games (Mayer 2020; Rahimi et al. 2021; Shute et al. 2021). One promising opportunity in digital GBL is harvesting and analysing learners' data generated in GBL environments for the purpose of understanding how learning happens in such learning environments to better support the learning process. GBL usually enables the generation of rich learning data since learners get

actively involved in the game (Hernández-Lara, Perera-Lluna, and Serradell-López 2019). Research shows that tracking, collecting, and analysing the data that reflects various learner interactions in GBL environments can offer valuable insights into how GBL helps learners to learn (e.g. Capatina et al. 2018; Chaudy and Connolly 2019; Hernández-Lara, Perera-Lluna, and Serradell-López 2019). For example, educators can monitor learners' progress during the gameplay and give real-time feedback on any recognised challenges (Capatina et al. 2018; Slimani et al. 2018) or they can see what kind of interaction patterns happen during GBL (Hernández-Lara, Perera-Lluna, and Serradell-López 2019; Huang et al. 2019). Such insights provided by data analytics open what has been a black box of learner engagement within educational games: where one could not see how learners perform during the gameplay, to what extent they interact with their peers or co-players, and what difficulties they usually run into in their learning process within GBL environments. Opening this black box has been referred to as *learning analytics*.

Learning analytics (LA) is broadly defined as 'the measurement, collection, analysis, and reporting of data about learners and their contexts, for purposes of understanding and optimizing learning and the environments in which it occurs' (Siemens and Long 2011, 34). In recent years, the use of LA to improve and optimise learners' learning in GBL contexts has received noticeable attention among scholars (e.g. Alonso-Fernández et al. 2020; Chaudy and Connolly 2019; Daghestani et al. 2020; Hernández-Lara, Perera-Lluna, and Serradell-López 2019). In GBL environments, which are highly interactive and data-rich (Alonso-Fernández et al. 2019b), there is an opportunity to produce massive user data (Slimani et al. 2018), which can be tracked, analysed, and visualised by LA to provide insights and evidence on the effects of different elements of the game on learning (Kang, Liu, and Qu 2017). Up-to-date, LA has been employed for various purposes in GBL. Some studies have utilised LA for assessment and evaluation, such as those conducted by Alonso-Fernández et al. (2019b), Chaudy and Connolly (2019), and Daoudi et al. (2021). On the other hand, other studies have focused on enhancing game adaptivity for effective learning, as seen in the works of Daghestani et al. (2020) and Harrison and Roberts (2014). Kang, Liu, and Qu (2017) utilised LA to uncover patterns of learners' learning behaviour in a GBL environment, while Hernández-Lara, Perera-Lluna, and Serradell-López (2019) adopted LA to explore what learners can learn from playing educational games.

At present, however, there is no comprehensive overview that provides a detailed understanding of the role

of LA in GBL practices, particularly in online learning environments within higher education contexts. It is not clear yet, to what extent, how, for what reason, and under what conditions LA is being used in GBL practices, especially in higher education contexts. We have found only two literature review studies that examined the relationships between LA and serious games (see Alonso-Fernandez et al. 2019a; Chaudy, Connolly, and Hainey 2014). Despite these literature review studies, there is still a lack of understanding regarding how the various elements of LA interact with each other in the context of serious games for two main reasons.

First, the systematic review study by Alonso-Fernandez et al. (2019a) mainly focuses on the purposes, algorithms, and recipients of data analytics in serious games. Similarly, the systematic review by Chaudy, Connolly, and Hainey (2014) offers a preliminary overview of the data types, techniques, and target users of LA tools in serious game contexts. While these studies provide valuable insights into the purposes, data, techniques, and target users of analytics in serious games, they did not discuss in-depth how these elements interact with each other and do not provide a framework that guides the use of data analytics in serious game contexts. Consequently, how the various elements of LA interact with each other in the context of serious games remains unclear. Therefore, there is a need for a more comprehensive study that explores the relationship between different elements of LA and GBL practices and proposes a framework that guides the use of LA in GBL contexts.

Second, the existing two review studies focused on serious games not so much on GBL. The distinction matters because serious games differ from GBL in terms of scope and objectives. Serious games exist in contexts beyond those of education, such as industries or businesses (Anastasiadis, Lampropoulos, and Siakas 2018). Hence, the objectives of serious games can be educational but not necessarily limited to educational objectives. On the other hand, GBL's scope solely is limited to the context of education (Connolly et al. 2012; Dehghanzadeh et al. 2021). Understanding the potential use of learning analytics in GBL is more aligned with the mission of LA since these environments are designed specifically for learning.

Finally, a large body of research on LA and GBL has been conducted in primary and secondary education (e.g. Huizenga et al. 2009; Partovi and Razavi 2019), however, there has been a growing interest in understanding the potential of LA in GBL in higher education contexts (e.g. Priyaadharshini, Dakshina, and Sandhya 2020). There are two important reasons why it is crucial to consider the role of LA in GBL specifically for higher education. From GBL's perspective, in primary

education, GBL often involves educational games designed to teach basic concepts (e.g. physics, mathematics, reading) and develop foundational skills (e.g. language learning). Also, games are often designed to be fun and engaging to hold the learners' attention and motivate them to learn (e.g. Hainey et al. 2016; Liu, Shaikh, and Gazizova 2020; Papastergiou 2009). While in higher education, GBL typically involves games designed to teach advanced concepts and skills. GBL in higher education, usually focuses on developing critical thinking, problem-solving, and decision-making skills that are usually relevant to learners' field of study. The games are often designed to be challenging and require a high level of cognitive engagement to complete (e.g. Amory and Seagram 2003). Another key difference between GBL in primary and higher education is the level of learner autonomy. In primary education, the teacher often directs the use of games and closely monitors learners' progress (e.g. Annetta et al. 2009; Ucus 2015). While, in higher education, learners are expected to take more responsibility for their own learning and may be given more freedom to explore and experiment within the game environment (e.g. Ebner and Holzinger 2007).

From LA's perspective, the use of LA in higher education and primary education tends to differ. While, the focus of LA in higher education is on supporting both teachers and learners, in primary education, LA seems to focus more on supporting teachers (Kovanovic, Mazziotti, and Lodge 2021). This is because learners in higher education are more independent and self-regulated compared to learners in primary education, who rely more on teacher guidance and support (Jossberger et al. 2010; Viberg, Khalil, and Baars 2020). Second, the goal and types of analytics used in LA could differ between higher education and primary education according to age and education level differences. In higher education, analytics may involve predictive modelling, clustering analysis, and social network analysis as the focus is on higher-order thinking skills (e.g. Banihashem, Noroozi et al. 2022), while in primary education, analytics may involve descriptive analysis, classification analysis, and clustering analysis (Ebner and Schön 2013).

Furthermore, while prior reviews did not differentiate in terms of mode (e.g. face-to-face, blended, or online learning) as the context in which GBLs have been used, the current review focuses specifically on online learning contexts in higher education. This is important to consider because the mode of education can impact the application of LA. For example, in face-to-face settings, the type of data that is collected is more observational through using cameras or other sensors (e.g. eye-tracking devices) that lead to more

qualitative data and different use of LA techniques (Ellis, Han, and Pardo 2017), while in online learning contexts, other types of data which are more quantitative regarding learners' learning activities can be tracked and stored by online tracking systems (Kew and Tasir 2021). That is to say that different modes of education can lead to different use of LA tools.

To summarise, current reviews are limited because they lack exploration of the impacts of various elements of LA in GBL practices within higher education settings. This review overcomes these issues by exploring the relationship between different elements of LA and GBL practices in higher education settings and proposes a framework that guides the use of LA in GBL contexts.

Therefore, the current systematic review enhances and enriches the literature in two primary ways. Firstly, by examining the role of LA in GBL practices within higher education contexts, the outcomes of this systematic review offer useful insights for educators on how LA can be utilised to enhance and facilitate GBL practices for learning outcomes. Secondly, as we investigate the critical elements of LA in supporting GBL, the findings of this review can aid in identifying research voids and guide future studies in this field.

The study is structured as follows. We begin by presenting the framework that guided our conceptualisation and research questions. We then introduce PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) (Moher et al. 2009) as the review method employed in this study, before presenting our findings for each research question and engaging in a thorough discussion of these results. To conclude our study, we offer insights into the implications of our findings for education and provide a summary of our conclusions.

2. Conceptual framework and research questions

In this study, we used a well-known LA Reference Model developed by Chatti et al. (2012) to frame and guide our study. This model provides an overview of the key components of LA systems and fosters our understanding of LA. According to this model, LA is composed of four main elements – the *what*, *how*, *why*, and *who*. (1) The *What* refers to the types of data that the system collects, analyzes, and reports. LA is a data-driven approach (Banihashem, Farrokhnia et al. 2022) and one of the key elements of any LA activity is to know what kind of data is generated and available (Chatti et al. 2012). In any LA system, data feed analytics and the richer data are available, the better insights can be provided by LA. According to Chatti

et al. (2012), there are two main sources to collect data including centralised education systems which refer to learning management systems (e.g. Moodle, Canvas, and Blackboard), and distributed learning environments which refer to data that comes from formal and informal personal learning environments (e.g. social media) (Chatti et al. 2012). (2) the *How* is about the methods used for performing analytics. LA adopts different techniques and methods to do analytics on data such as clustering, classification, data mining, information visualisation, and social network analysis (Avella et al. 2016; Chatti et al. 2012). The adoption of LA techniques and methods is highly dependent on the objectives of the analytics task (Chatti et al. 2012). (3) The *Why* refers to the objectives and reasons why the system captures, analyzes, and reports data. LA is used for many reasons in educational settings. For example, LA can be used for assessing and monitoring learners' learning progress or it can be used for providing scaled personalised feedback (Banihashem, Noroozi et al. 2022; Pardo et al. 2019). According to Chatti et al.'s model (2012), the main reasons to use LA include monitoring, analysis, prediction, intervention, tutoring/mentoring, assessment, feedback, adaptation, personalisation, recommendation, and reflection. (4) And the fourth element of the LA Reference Model relates to the *Who*. That is, who the recipients of the analytics are (Chatti et al. 2012). According to Chatti et al.'s model (2012), depending on the purpose of the use of LA in educational settings, different stakeholders including learners, teachers, (intelligent) tutors/mentors, educational institutions (administrators and faculty decision-makers), researchers, and system designers can benefit from LA. The four elements of the LA Reference Model (Chatti et al. 2012) are supported by the results of other relevant studies (see Ferguson 2012; Greller and Drachslar 2012).

Each of the four components of the LA Reference Model (Chatti et al. 2012) is crucial to consider in a review. In terms of the first component, data (*what*) are the core foundation of LA since LA is a data-driven field of study (Banihashem and Macfadyen 2021). There is a need to investigate and map the literature on what data is being used by LA to support online GBL in higher education. In terms of the second component, methods (*how*), LA can leverage an extensive range of analytical algorithms, techniques, and methods such as statistics, visualisations, data mining, and social network analysis to analyse captured data (Chatti et al. 2012; Greller and Drachslar 2012). LA's techniques and methods can differ from one application to another based on the types of data and the goal of LA. For example, social network analysis techniques can be

applied if interactive data are collected and if the goal is to explore the interaction between nodes (De Laat et al. 2007; Saqr et al. 2018). There is currently no clear picture of LA's techniques and methods used to support online GBL in higher education. It is also not clear in the literature at what levels of analytics (descriptive, diagnostic, predictive, and prescriptive) (Maoz, 2013) different LA techniques and methods can be used. For example, we do not have an overview at which levels social network analysis methods can be used. Are social network analysis methods only used at the descriptive level for describing the connections between the nodes or are they used at the predictive level for predicting the value of an attribute based on the value of other attributes (Turkington et al. 2018)? The current review aims to clarify this matter. In terms of the third component, objectives (*why*), there are many different reasons why LA is used in education including monitoring, reflection, prediction, intervention, feedback, personalisation, adaptation, and recommendation. A review of the literature also confirms these objectives for LA (see Avella et al. 2016; Banihashem, Noroozi et al. 2022; El Alfy, Gómez, and Dani 2019; Greller and Drachsler 2012). For all of these reasons, there is a need to further clarify the reasons why LA has been used in online GBL at the higher education level. Finally, in terms of the fourth component, the recipients (*who*), different people may benefit from the integration of LA in education. Chatti et al. (2012) indicate that stakeholders of LA in education may include learners, educators, tutors, mentors, institutions, researchers, and system designers. This finding is supported by prior studies (e.g. Greller and Drachsler 2012). Beneficiaries of LA can differ based on the reasons why LA is used. For example, if LA is used for the personalisation of learning, then learners are expected to be the main stakeholders. In the case of using LA in online GBL at a higher education level, it is not clear yet who are the main stakeholders of LA applications.

Therefore, in this study, we investigate and provide an overview of these four main elements of LA in the context of online GBL in higher education. We believe that this systematic review could contribute to the LA and GBL research community by providing a framework on what, why, how, and who should be taken into consideration when it comes to applying LA to support online GBL in higher education. Thus, drawing on the conceptual framework provided above, the following research questions are formulated to guide this systematic review study:

RQ1. What types of data are used by LA to support online GBL in higher education?

RQ2. What methods are used by LA to support online GBL in higher education?

RQ3. For what reasons is LA used in online GBL in higher education?

RQ4. Who are the stakeholders of using LA in online GBL in higher education?

3. Method

In this study, we developed a search strategy based on the PRISMA methodology (Moher et al. 2009) to define the inclusion criteria and identify the relevant studies. PRISMA is a widely recognised guideline for conducting and reporting systematic reviews. This method includes a checklist that helps with transparent review and recommends how a systematic review should be reported (Page et al. 2021).

3.1. Developing the search strategy

The process of conducting a search strategy for finding relevant studies included two components. First, the keywords that cover the main body of the literature were defined based on the review of prior systematic studies in the field of LA and GBL (e.g. Alonso-Fernandez et al. 2019a). The keywords were ('LA' OR analytic* OR 'data mining') AND (gamif* OR 'computer games' OR 'online games' OR 'mobile games' OR 'digital games' OR 'video games' OR 'simulation games' OR 'massively multiplayer online game' OR MMOG OR 'serious games') AND (learning OR education OR teaching OR instruction). It should be noted here that we included 'serious games' as one of the keywords since, as we said, it covers a broader scope and this keyword may be used for a study with a learning purpose. Later in the screening phases, we included only those serious game studies that were conducted solely for learning purposes within higher education contexts.

Second, our search queries primarily focused on Web of Science, Scopus, ERIC, IEEE, and Proceedings of the International Conference on Learning Analytics and Knowledge (LAK). These databases were chosen because they are known to contain a vast amount of literature on LA and GBL.

3.2. Defining inclusion and exclusion criteria

In this study, inclusion criteria were defined in two phases. In the first phase, for the initial screening of the relevant literature, we decided to (1) include only publications in English since the vast majority of

scientific papers are published in English; (2) consider only publications between 2011 to 2021 since the first Learning Analytics and Knowledge (LAK) conference was held by SoLAR (Society for Learning Analytics and Research) in 2011 and most LA publications were published after 2011 (Siemens and Gašević 2012); and (3) include only peer-reviewed papers because they carry an acceptable level of originality, validity, and reliability as reviewed by academicians. Therefore, other types of documents such as non-peer-reviewed papers, government reports, unpublished manuscripts, editorials, book chapters, theses, dissertations, and author notes were excluded from this review.

After a preliminary review of the literature according to the first phase inclusion criteria, in the second phase, four other inclusion criteria were employed to screen identified publications and narrow down studies to match the scope of this study. These four defined criteria were as follows: (1) Only empirical studies were considered for this study and publications with conceptual and review nature were excluded. This decision was made because the results of empirical studies are scientifically reliable to address the research questions; (2) the learning context of the studies was limited to online learning environments since this study seeks to address the effects of LA applications solely in online GBL; (3) the education level of the studies was restricted to higher education. Therefore, studies with other education levels like primary and secondary education were excluded from further analysis; and (4) only publications with clear findings and implications that could address at least one of our research questions were included. This means that the findings of the included publication must either entail information regarding the LA data and methods used in the GBL context (what and how) or the reasons why LA is used or information regarding who was the recipient of LA services in GBL contexts. To do this, the authors decided to read titles, abstracts, and, in cases where it was necessary, the whole full text. Studies that did not meet the above-mentioned criteria through screening were excluded from further analysis.

3.3. Identifying relevant studies

The search query was performed in the selected databases with defined keywords based on the first phase inclusion criteria. The search results led to 484 publications in total and the result for each database reported as follows: Web of Science ($n = 108$), Scopus ($n = 256$), ERIC ($n = 69$), IEEE ($n = 35$), and LAK Conferences ($n = 16$). After a preliminary screening of the identified publications, 133 studies were removed since they were

recognised as duplicates, and 351 publications remained. In the next step, the remaining publications were screened according to the second phase inclusion criteria. After screening the titles and abstracts, 68 articles were deleted since they were off-topic. In addition, 52 papers were excluded since they were recognised as conceptual and review papers, not empirical studies. At this point, 231 studies remained for full-text screening since we could not address inclusion criteria by reading the abstract. Our full-text review showed that 84 studies were related to GBL but not to LA, 81 publications were not conducted in higher education contexts, 35 papers were found to be non-empirical studies, 6 papers were removed since they were related to LA but not to GBL, and 5 papers were found not to be within an online learning environment. Finally, the full-text screening left us with 20 publications for quality appraisal.

3.4. Quality appraisal

We adopted the quality appraisal criteria (Theelen, Van den Beemt, and den Brok 2019) to critically review the quality of the remaining publications. The criteria for quality appraisal consisted of two sections: one for critical evaluation of the qualitative studies (9 items) that was based on the work of Savin-Baden and Major (2007) and the other for quality appraisal of quantitative studies (7 items) that was built on a checklist developed by NICE (2012). The criteria for quality appraisal are scored from no mention (0) to extensive mention (3). As suggested by Theelen, Van den Beemt, and den Brok (2019), it was decided to exclude studies whose mean score was less than 2 points. Quality appraisal was conducted on the remaining quantitative ($n = 16$), qualitative ($n = 1$), and mixed studies ($n = 3$), and the results showed that all the studies met methodologically the minimum score for inclusion and were therefore included in the final analysis (Table 1). The flowchart of the selection process is illustrated in Figure 1.

3.5. Included publications

As mentioned above, the screening of the identified publications left us with 20 studies for the final analysis. The details of the selected studies are provided in Table 2.

3.6. Analysis strategy

A coding scheme was developed (Table 3) based on Chatti et al.'s (2012) LA framework to analyse selected studies. This coding scheme entailed four categories in

Table 1. Quality appraisal criteria.

Criteria	Not mention (0)	Some mention (1)	Good mention (2)	Extensive mention (3)
<i>Criteria for qualitative studies</i>				
Study methodologically is clear		1	2	1
Study theoretically substantiated		1	2	1
Ethical process transparent	2	1		1
Researcher(s) relation to participants are clear		1	2	1
Researchers(s) relation to the data are clear		1	2	1
Researcher(s) take a critical stance towards own research			3	1
Congruence between methodology and methods used for data collection, analysis, and interpretation		1	1	2
Participant involvement in data interpretation			2	2
Limitations voiced	1		2	1
<i>Criteria for quantitative studies</i>				
Is the source population or source area well-described?		1	13	5
Were interventions and comparisons well-described?		1	10	8
Were outcome measures reliable?		1	7	11
Were outcomes relevant?			9	10
Were the analytical methods appropriate?		1	10	8
Are the study results internally valid (i.e. unbiased)?		1	12	6
Are the findings generalisable to the source population?		2	7	10

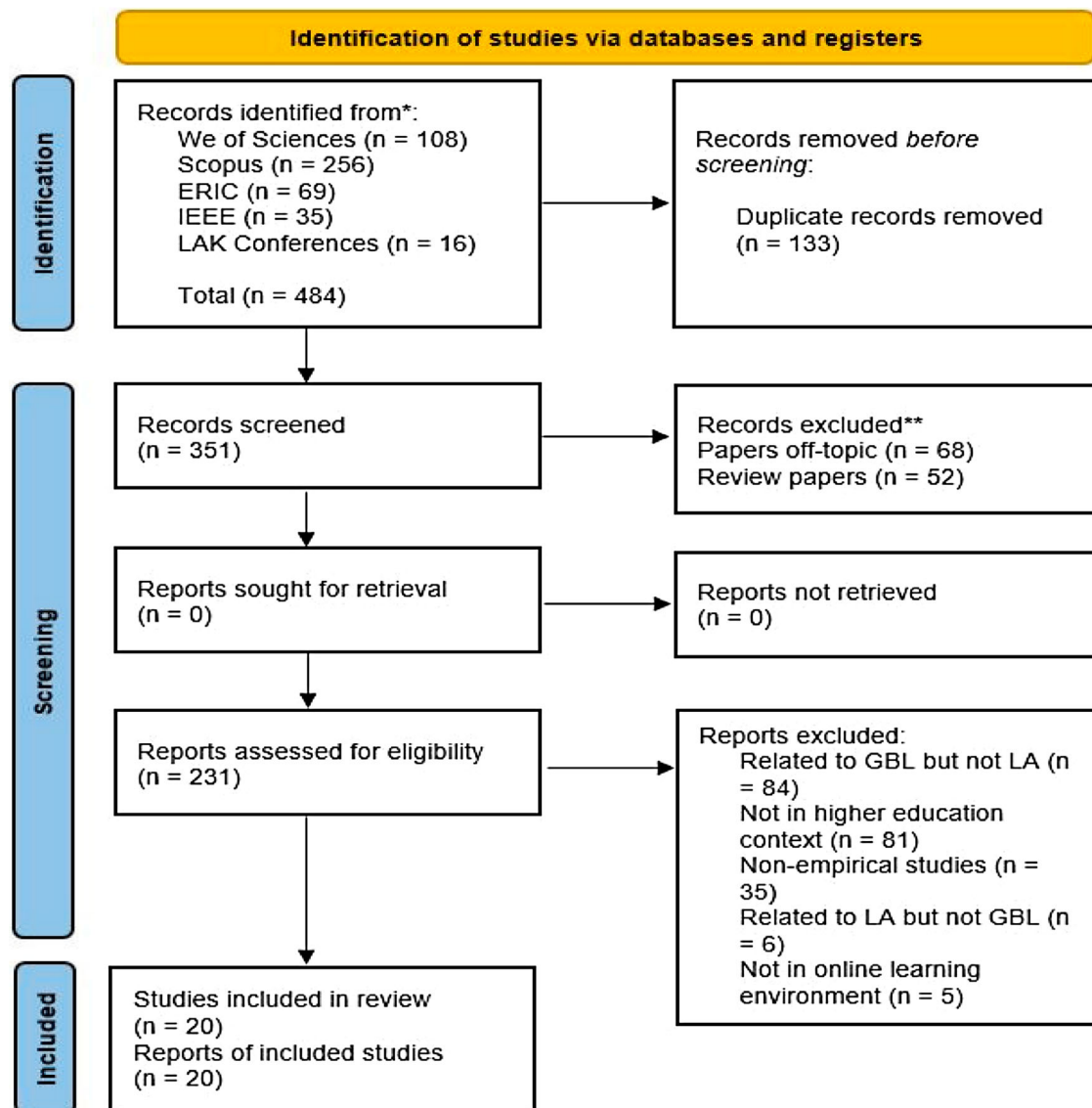
**Figure 1.** Flowchart of the selection process.

Table 2. Characteristics of the selected studies.

Authors (year)	Journal/Conference	Country of study	Type of study	Content of the context	Learning environment
Capatina et al. (2018)	<i>Behaviour & Information Technology</i>	Italy, France, Romania	Qualitative	Management	Simbound
Cengiz et al. (2017)	<i>The International Journal of Engineering Education</i>	Turkey	Quantitative	Computer Science	Mine4Escape
Chaudy and Connolly (2019)	<i>Entertainment Computing</i>	The UK	Quantitative	Programming	EngAGe
Daghestani et al. (2020)	<i>Computer Applications in Engineering Education</i>	Egypt	Quantitative	Computer Science	AGLS
Daoudi et al. (2021)	<i>Entertainment Computing</i>	France	Quantitative	Management	iScen
Forsyth, Graesser, and Millis (2020)	<i>Technology, Knowledge, and Learning</i>	The USA	Quantitative	Research methods/ Psychology	Operation ARA
Gañán et al. (2017)	<i>International Journal of Web Information Systems</i>	Spain	Quantitative	N/A	ICT-FLAG
Hernández-Lara, Perera-Lluna, and Serradell-López (2019)	<i>Computers in Human Behavior</i>	Spain	Quantitative	Business	Cesim Global Challenge
Huang et al. (2019)	<i>Distance Education</i>	China	Quantitative	Management	Moodle
Konstantakopoulos et al. (2019)	<i>Applied Energy</i>	The USA	Quantitative	Energy saving	Social Game
Kuo and Chuang (2016)	<i>Computers in Human Behavior</i>	Taiwan	Mixed	Academic knowledge	LOPUPA
Labonte-LeMoyne et al. (2017)	<i>Business Process Management Journal</i>	Canada	Quantitative	Technology and pedagogy innovations	ERPsim
Loh, Li, and Sheng (2016)	<i>Computers in Human Behavior</i>	The USA	Quantitative	N/A	Course of Actions (COAs)
Minović et al. (2015)	<i>Computers in Human Behavior</i>	Spain	mixed	Computer Science	EduGameLO
Muñoz-Merino et al. (2017)	<i>Computer Applications in Engineering Education</i>	Spain	Quantitative	Physics	Khan Academy
Palomo-Duarte et al. (2019)	<i>Journal of Gaming & Virtual Worlds</i>	Spain	Quantitative	German language	The Hidden Room
Rodríguez-Cerezo et al. (2014)	<i>Computers in Human Behavior</i>	Spain	Mixed	Computer Science	Evaluators
Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017)	<i>International Journal of Engineering Education</i>	Spain	Quantitative	Engineering	Khan Academy
Slimani et al. (2018)	<i>International Journal of Emerging Technologies in Learning</i>	Morocco	Quantitative	Biology	ELISA
Taub and Azevedo (2018)	<i>Journal of Educational Data Mining</i>	The USA	Quantitative	Microbiology	Crystal Island

line with the research questions including types of data (what), methods (how), objectives (why), and stakeholders (who). The software called ATLAS.ti 9 (Friesen

2019) was used to code and analyse data. ATLAS.ti is a powerful tool for researchers to manage and analyse large amounts of qualitative data. We used this tool

Table 3. The coding scheme to analyse included publications.

Category	Code	Label	Definition
Types of data (what)	Type of data	A1	Refers to the types and nature of data such as learners' information data, academic data, performance data, interaction data, etc. that are collected by LA tools to GBL
Methods and techniques (how)	Visualisations	B1	Refers to the information visualisation methods and techniques that are used by LA to analyse LA data in GBL contexts
	Data mining	B2	Refers to the data mining methods and techniques that are used by LA to analyse LA data in GBL contexts
	Social network analysis	B3	Refers to the social network analysis methods and techniques that are used by LA to analyse LA data in GBL contexts
	Text mining	B4	Refers to the text mining methods and techniques that are used by LA to analyse LA data in GBL contexts
Objectives (why)	Monitoring	C1	Refers to the monitoring goal of LA in GBL contexts
	Prediction	C2	Refers to the prediction goal of LA in GBL contexts
	Tutoring	C3	Refers to the tutoring goal of LA in GBL contexts
	Assessment	C4	Refers to the assessment goal of LA in GBL contexts
	Adaptation	C5	Refers to the adaptation goal of LA in GBL contexts
	Personalisation	C6	Refers to the personalisation goal of LA in GBL contexts
	Recommendation	C7	Refers to the recommendation goal of LA in GBL contexts
	Reflection	C8	Refers to the reflection goal of LA in GBL contexts
	other	C9	Refers to any other goal of LA in GBL contexts
Stakeholders (who)	Learners	D1	Refers to the learners as the stakeholders of LA's use in GBL contexts
	Educators	D2	Refers to the educators as the stakeholder of LA's use in GBL contexts
	Tutors/mentors	D3	Refers to the tutors/mentors as the stakeholders of LA's use in GBL contexts
	Institutions	D4	Refers to the institutions as the stakeholder of LA's use in GBL contexts
	Researchers	D5	Refers to the researchers as the stakeholders of LA's use in GBL contexts
	System designers	D6	Refers to the system designers as the stakeholders of LA's use in GBL contexts
	Course designers	D7	Refers to the course designers as the stakeholders of LA's use in GBL contexts
	Other	D8	Refers to any other group of people as the stakeholder of LA's use in GBL contexts

for data analysis because it is user-friendly and provides a variety of features to help users organise and analyse their data, including tools for coding and tagging data, creating memos and notes, and visualising relationships between different pieces of data. It also offers advanced search and retrieval capabilities, allowing users to quickly locate specific information within their datasets (Friese 2019; Hwang 2008). To determine the inter-rater reliability for analysing the included publications, three articles were randomly selected and analysed by the two authors (coders) and the inter-rater reliability between coders reached 95.2 percent agreement ($\kappa = .83$, $p < .001$) indicating high consistency for the coding scheme.

4. Results

4.1. RQ1. What types of data are used by LA to support online GBL practices in higher education?

Data analysis for the first research question resulted in a total of 204 codes describing the types of data used by LA to support online GBL in higher education. We classified the collected data into four distinct categories, which are gameplay data (internal and external), assessment data, player data, and game metadata. We further elaborated on each category to provide a clear understanding of which data falls under which category (see Table 4). Below, these categories are further discussed.

4.1.1. Gameplay data

This group of data was found to be the dominant data type used by LA to support online GBL in higher education as it was identified in all included publications (N of studies = 20, 100%) and entailed 51% of the total codes (N of codes = 103). Gameplay data refers to the kinds of data that were being collected during the gameplay. According to reviewed studies, gameplay data collection takes place either internally or externally. Internal gameplay data refers to data collected during the gameplay from the game software itself such as time of play (Slimani et al. 2018), the number of taken badges (Daghestani et al. 2020), learner-game interaction data (Loh, Li, and Sheng 2016), and received points (Konstantakopoulos et al. 2019). These internal gameplay data were incorporated in 100% of the studies (N of studies = 20) and entailed 38% of the total codes (N of codes = 77). External gameplay data refers to when the data are collected during the game externally outside of the game software such as players' eye-tracking data, emotional and cognitive reactions during the gameplay (Taub and Azevedo 2018), and some other

visual data (e.g. real-time video recordings from webcams) (Daoudi et al. 2021). External gameplay data were collected in 30% of the studies (N of studies = 6) and accounted for 13% of the codes (N of codes = 26). Since gameplay data are generated during a learning activity, such data can provide information and reflections on how players are doing and inform the players about their learning process during gameplay (Daoudi et al. 2021). In addition, gameplay data can help educators with tracking and monitoring learners' learning activities to make better pedagogical decisions (Minović et al., 2015).

4.1.2. Assessment data

This group of data includes varied types of data that can be obtained from quizzes and tests outside the game (e.g. measures of learning collected outside of the game and outside the time of gameplay). Assessment data were collected in 80% of the studies (N of studies = 16) and accounted for 36% of the codes (N of codes = 74). This data type can be used by educators for effective formative and summative learner evaluation (Chaudy and Connolly 2019; Huang et al. 2019). In addition, assessment data contain data from interviews, open-ended questions, surveys, and scales such as the critical thinking scale (Huang et al. 2019). This kind of data can serve as supplementary data for those data that are mined from LMS or game platforms (Cengiz et al. 2017; Huang et al. 2019). For example, the Student Engagement Scale (SES) was used alongside the gameplay data for exploring learners' engagement in Daghestani et al. (2020). A review of the used survey data in the selected GBL studies revealed that this data type can entail high reliability and validity since the instruments used for collecting the data may have already been validated in earlier research. For example, the Student Engagement Scale (SES) used by Daghestani et al. (2020) has already been validated in another study by Dixon (2015).

4.1.3. Player data

This kind of data refers to the data that relates to the player himself/herself. In general, player data can tell us who are the players and from what kinds of educational and psychological background and features, they are coming. Player data were collected in 30% of the studies (N of studies = 6) and accounted for 9% of the codes (N of codes = 20). Player data include varied data types such as demographic data (e.g. age, gender, language) (Daghestani et al. 2020; Taub and Azevedo 2018) which is one of the most commonly employed types of data in LA studies (Tempelaar, Rienties, and Giesbers 2015), learners' academic background data

Table 4. Types of data used by LA to support online GBL in higher education.

Category		Types of data	N of codes* (%)	N of Studies (%)	References
Gameplay data	Internal	Engagement as measured by gameplay, time spent on the game, number of attempts, time of attempts, learner logs, type of interaction, level of interaction, time-spent on interaction, number of quests, number of taken badges, number of received rewards, number of the leaderboard views, time-spent on navigation, interaction with the platform, interaction with other characters, time spent on task, interacting with characters, e-text, narratives, number of words used by learners to answer questions in the game, submissions of the solutions, received points, total time, number of received gift cards, received ranking, browsing data, time-series data, number of pages viewed, average duration, brainstorm activity, sales revenues, quantity sold, types of products, duration of use, action data, watching videos in the game, video views in the game, number of completed videos in the game, time of play, last level achieved, number of players' clicks, players experience data, trace data, log file data, score-points, exercises, number of correct answers, number of false answers, number of completed challenges, number of completed levels	77 (38%)	20 (100%)	Capatina et al. (2018), Cengiz et al. (2017), Chaudy and Connolly (2019), Daghestani et al. (2020), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Gañán et al. (2017), Hernandez-Lara et al. (2019), Huang et al. (2019), Konstantakopoulos et al. (2019), Kuo and Chuang (2016), Labonte-LeMoyne et al. (2017), Loh, Li, and Sheng (2016), Minović et al. (2015), Muñoz-Merino et al. (2017), Palomo-Duarte et al. (2019), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018), Taub and Azevedo (2018)
	External	Emotional reactions, cognitive reactions, behavioural reactions, communications, facial expressions of emotions, eye-tracking data, number of posts, number of messages sent, visual data (real-time video recordings from webcams), video tutorials, exchanged messages during playing, recorded reactions, learners' posts, discussion forum chat	26 (13%)	6 (30%)	Chaudy and Connolly (2019), Daghestani et al. (2020), Daoudi et al. (2021), Huang et al. (2019), Hernandez-Lara et al. (2019), Taub and Azevedo (2018)
Assessment data		Quizzes, grades, tests, multiple-choice tests, learners' engagement scale, game engagement questionnaire, critical thinking evaluation scale, online survey, self-report questionnaire, self-assessment scale, web questionnaire, open-ended questions, interviews, qualitative questions, system usability scale, perceived interest questionnaire, computer emotion scale, pre-tests, post-tests, feedback (correct/incorrect), assignments, grammatical errors	74 (36%)	16 (80%)	Cengiz et al. (2017), Chaudy and Connolly (2019), Daghestani et al. (2020), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Gañán et al. (2017), Huang et al. (2019), Kuo and Chuang (2016), Konstantakopoulos et al. (2019), Kuo and Chuang (2016), Labonte-LeMoyne et al. (2017), Minović et al. (2015), Muñoz-Merino et al. (2017), Rodríguez-Cerezo et al. (2014), Slimani et al. (2018), Taub and Azevedo (2018)
Player data		Demographics, player characteristics, learner data, age, language, gender, ID number, personal data, academic background, educational level, number of players (learners), educational user data, learners' profile, psychological features	20 (9%)	6 (30%)	Cengiz et al. (2017), Chaudy and Connolly (2019), Daghestani et al. (2020), Daoudi et al. (2021), Labonte-LeMoyne et al. (2017), Taub and Azevedo (2018)
Game metadata		History of e-learning systems, history of game, game genre, game story	7 (4%)	4 (20%)	Chaudy and Connolly (2019), Daghestani et al. (2020), Gañán et al. (2017), Labonte-LeMoyne et al. (2017)

*Code (Types of data identified from reviewed studies) – N of codes (the occurrence of the types of data in reviewed studies).

(Chaudy and Connolly 2019), education level (Cengiz et al. 2017), and player's profile (the emotional profiles which are detected based on formative and summative evaluation of players' produced emotional data) (Daoudi et al. 2021). In addition, player data include psychological data that are rooted in learners' psychological features and behaviours such as learners' emotional, cognitive, and behavioural reactions (Daoudi et al. 2021). This type of data can help with the interpretation of the players' emotional responses

during the gameplay (Daoudi et al. 2021; Taub and Azevedo 2018). This player data can help to understand why learners experience such emotions in the gameplay (Daoudi et al. 2021; Taub and Azevedo 2018). Moreover, player data can be used for different reasons alongside other data types in GBL practices. For example, learners' demographic data and learners' profile data were used alongside the gameplay data to provide an adapted GBL environment (Daghestani et al. 2020) to better assess learners' performance improvement

(Slimani et al. 2018) or to explore learners' emotions in the game (Daoudi et al. 2021).

4.1.4. Game metadata

This data type refers to any kind of data that can be collected before gameplay about the game itself (and not about the player), including data regarding the history of e-learning systems or game platforms (Daghestani et al. 2020) and game genre and story (Chaudy and Connolly 2019). Game metadata was utilised in 20% of the studies (N of studies = 4) and accounted for 4% of the codes (N of codes = 7). These data can be useful to understand in what context the game is played. This kind of LA data might be useful when educators want to interpret the LA findings and learners' performance in the game. For example, in a study, game metadata can be used for better interpretation of learners' performance assessments (Gañán et al. 2017).

4.2. RQ2. What methods are used by LA to support online GBL practices in higher education?

We identified 73 codes from the selected studies in terms of LA methods employed. These methods were grouped into categories including data mining, visualisations, text mining, and social network analysis methods (Table 5).

4.2.1. Data mining

The number of codes identified for LA methods in the selected online GBL studies indicates that data mining methods are the most commonly used group of methods to do analytics on collected LA data (N of codes = 41, 56%; N of studies = 14, 70%). Generally, data mining methods are used to discover different forms of information such as patterns, correlations, and anomalies in a large set of data (Lee and Siau 2001). Some important data mining methods used to support online GBL in higher education include association rule mining (Cengiz et al. 2017), classification (Daoudi et al. 2021; Loh, Li, and Sheng 2016), regression (Daghestani et al. 2020), clustering (Palomo-Duarte et al. 2019; Slimani et al. 2018), predictive modelling (Daghestani et al. 2020; Forsyth, Graesser, and Millis 2020), relationship mining (Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos 2017), and sequential pattern mining (Taub and Azevedo 2018). A review of the selected GBL studies suggests that data mining methods can be used for tracking player's performance during gameplay (Capatina et al. 2018), providing intelligent predictive feedback (Cengiz et al. 2017; Gañán et al. 2017), detecting and clustering players based on their performance and activities (Palomo-Duarte et al. 2019; Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos 2017), and analysing metacognitive monitoring (Taub and Azevedo 2018).

Table 5. Methods used by LA to support online GBL in higher education.

Category	Methods	N of codes* (%)	N of Studies (%)	References
Data mining	Association rule mining, classification, regression, clustering, prediction, the distillation of data, decision tree, naive Bayesian, k-nearest neighbour, fuzzy-set qualitative comparative analysis, information trails, deep learning, feature distillation, k-means, relationship mining, sequential pattern mining, differential sequence mining, linear discriminant analysis, subsequent analysis, partial least squares discriminant analysis, lossy approach, Ward's method, EM algorithms, exploratory data analysis	41 (56%)	14 (70%)	Capatina et al. (2018), Cengiz et al. (2017), Daghestani et al. (2020), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Kuo and Chuang (2016), Konstantakopoulos et al. (2019), Loh, Li, and Sheng (2016), Palomo-Duarte et al. (2019), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018), Taub and Azevedo (2018)
Visualisations	Time-series graphs, tree-based methods, graphical virtual feedback, multiple graphs and tables, pivot tables, simulations, line plots, bar charts, facial expressions, word clouds, pie charts, network diagrams, box plots, scatter plots, histogram plots, dashboards, flow charts	21 (29%)	17 (85%)	Capatina et al. (2018), Daghestani et al. (2020), Daoudi et al. (2021), Huang et al. (2019), Kuo and Chuang (2016), Labonte-LeMoyne et al. (2017), Gañán et al. (2017), Konstantakopoulos et al. (2019), Huang et al. (2019), Loh, Li, and Sheng (2016), Minović et al. (2015), Muñoz-Merino et al. (2017), Palomo-Duarte et al. (2019), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018), Taub and Azevedo (2018)
Text mining	Natural language processing, syntactic-semantic analysis, content analysis, text analytics	7 (10%)	4 (20%)	Daoudi et al. (2021), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Huang et al. (2019), Rodríguez-Cerezo et al. (2014)
Social network analysis	Neural network, social network analysis	4 (5%)	2 (10%)	Konstantakopoulos et al. (2019), Huang et al. (2019)

*Code (Methods identified from reviewed studies) – N of codes (the occurrence of the methods in reviewed studies)

4.2.2. Visualisations

Visualisation methods were found in 85% of the studies (N of studies = 17) and accounted for 29% of the codes (N of codes = 21). Visualisation methods refer to the representation of data in a visual format (Khan and Khan 2011). As our review revealed, visualisation methods are mentioned in the selected studies more than any other LA methods (N of studies = 17, 85%). This can be explained by the LA literature where it is said that the majority of LA reports are presented in a visualised way through an LA dashboard by using different forms of data representations such as charts, tables, graphs, and diagrams (Chatti et al. 2012; Guerra et al. 2020; Verbert et al. 2013). Although visualisation methods were found to be more prevalent than data mining methods among selected studies, the number of data mining methods used in the selected studies was higher than visualisation methods. This means that more codes were identified for data mining methods in reviewed publications. For example, in the Daghestani et al. (2020) study only two codes for visualisation methods (bar charts and flow charts) were identified, while the number of identified codes for data mining methods was ten (e.g. decision tree, prediction, clustering, classification). In the selected GBL studies, different data visualisation methods such as time-series graph portraying (Capatina et al. 2018), tree-based methods (Daoudi et al. 2021), simulations (Minović et al., 2015), and network diagrams (Huang et al. 2019) are used to represent data in diverse visual ways. The selected GBL studies show that visualisation methods are used for different reasons such as monitoring (Capatina et al. 2018), adaptation (Daghestani et al. 2020), assessment (Daoudi et al. 2021), decision-making (Loh, Li, and Sheng 2016), and visualisation of the learning process (Minović et al., 2015) in online GBL within higher education contexts.

4.2.3. Text mining

Text mining methods, the process of drawing meaning from a text, were identified in 20% of the studies (N of studies = 4) and accounted for 10% of the codes (N of codes = 7). Text mining is generally used to categorise, cluster, summarise, and extract concepts from written texts (Fiaidhi 2014). In the included GBL studies, different methods of text mining such as natural language processing (Hernández-Lara, Perera-Lluna, and Serradell-López 2019), semantic analysis (Rodríguez-Cerezo et al. 2014), content analysis (Huang et al. 2019), and text analytics (Hernández-Lara, Perera-Lluna, and Serradell-López 2019) are used. Reviewed publications indicate that text mining methods are mostly used to provide insights into learners' interactions during gameplay (e.g. Hernández-

Table 6. Categorisation of the LA methods based on the level of analytics (Maoz, 2013).

LA methods	LA levels			
	Descriptive	Diagnostic	Predictive	Prescriptive
Data mining			*	*
Text mining	*	*		
Social network analysis	*	*		
Visualisation	*	*	*	

Lara, Perera-Lluna, and Serradell-López 2019; Huang et al. 2019).

4.2.4. Social network analysis

In our pool of selected studies, only 10% of the studies (N of studies = 2) and 5% of the codes (N of codes = 4) referred to social network analysis. This analytics method is defined as a process of studying relationships, interactions, and communications between two or more nodes (Saqr and Alamro 2019). While, in social network analysis, nodes can be persons, communities, content, or anything else (Streeter and Gillespie 1993), in the selected GBL studies, nodes typically refer to players (learners) (Huang et al. 2019). Reviewed publications suggest that social network analysis methods are commonly used to discover interaction patterns (e.g. Huang et al. 2019; Konstantakopoulos et al. 2019). For example, in one study, Huang et al. (2019) adopted the social network analysis method to analyse learners' online interactive patterns (network size, density, and degree centralisation) and posting quantity in a GBL environment.

As presented in Table 6, the analytics in LA can be divided into four main levels including descriptive, diagnostic, predictive, and prescriptive (Eriksson, Bigi, and Bonera 2020; Henry and Venkatraman 2015; Maoz, 2013; Turkington et al. 2018). In the following paragraphs, we explain the levels of analytics and discuss how LA methods can be used at different levels.

In general, descriptive analytics as the primary level of analytics reports information in a visual way about what has happened in the past or what is happening now (Henry and Venkatraman 2015). In other words, in descriptive analytics, LA methods are used to describe and provide a summary of how learners are doing. Although all LA methods can potentially be used for descriptive analytics purposes, visualisation, text analytics, and social network analysis methods tend to be used more often since these methods are generally used for describing the activities that have been done. Diagnostic analytics is a form of analytics that tries to answer the question 'why did it

happen?’ (Maoz, 2013). This level of analytics takes steps further than just simply describing the ‘past events’ and highlights differences and similarities in the findings (Eriksson, Bigi, and Bonera 2020). Diagnostic analytics can be used by educators to identify learners who are outlying from the mainstream of learning (Turkington et al. 2018). Similar to descriptive analytics, it is expected that visualisation, text analytics, and social network analysis methods are mainly applied for diagnostic analytics. For example, social network analysis methods are used at a diagnostic level to see which learner’s interaction performance outlies other learners (Huang et al. 2019). Predictive analytics normally tells us ‘what will happen’ (Maoz, 2013). Predictive analytics helps us with predicting the value of an attribute based on the value of other attributes (Turkington et al. 2018). In this level of analytics, typically, data mining methods such as classification, regression, clustering, support vector machines, and decision tree methods are used to model and predict learners’ learning behaviour patterns (Maoz, 2013; Patil, Nemade, and Soni 2018). Alongside data mining methods, at the predictive analytics level, visualisation methods are used to report data mining results in a visual way for LA users. Finally, LA methods can be adopted at a prescriptive analytics level where analytics advise and suggest different decisions, options, actions, and interventions toward a solution based on the results reported in previous LA levels (Turkington et al. 2018). The educational recommender systems can be a good example of LA at the prescriptive analytics level (Bodily and Verbert 2017). Similar to predictive analytics, data mining methods tend to be more applied at this level of analytics.

4.3. RQ3. For what reasons are LA used in online GBL studies in higher education?

In total, we identified 104 codes from the included publications that represent the purpose of LA used in online GBL within higher education contexts. The objectives we found for LA in GBL studies have gone beyond our initial expectations based on the suggested theoretical model. What we found is that LA purposes in GBL studies are varied from assessment (Chaudy and Connolly 2019) to self-regulation (Taub and Azevedo 2018). The intensity and prevalence of these services, however, are not similar. While, according to the selected literature, assessment is deemed to be one of the main LA objectives in GBL studies, self-regulation is mentioned in only one study. Since our findings for LA objectives vary from anticipated objectives to additional objectives, therefore, we decided to categorise our findings into two main categories, including anticipated objectives based on our theoretical model which entail 62 codes (Table 7), and additional objectives which consist of 42 codes (Table 8). This categorisation helps with the illustration of the distinction between the objectives we expected from our theoretical model and other additional objectives that we found from the selected literature.

4.3.1. Assessment

Assessment as an LA objective was found in 35% of studies (N of studies = 7) and 34% of the codes (N = 21) accounted for that purpose. According to the literature, providing assessment services is one of the key purposes followed by LA studies over the years (e.g. Knight, Shum, and Littleton 2014; 2020). Even in some studies, LA is considered an assessment tool

Table 7. Anticipated objectives of LA use in online GBL in higher education.

Category	Objectives	N of codes* (%)	N of Studies (%)	References
Anticipated objectives	Assessment	21 (34%)	7 (35%)	Chaudy and Connolly (2019), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Gañán et al. (2017), Huang et al. (2019), Konstantakopoulos et al. (2019), Rodríguez-Cerezo et al. (2014)
	Monitoring	12 (19%)	8 (40%)	Capatina et al. (2018), Chaudy and Connolly (2019), Daghestani et al. (2020), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Huang et al. (2019), Konstantakopoulos et al. (2019), Minović et al. (2015), Taub and Azevedo (2018)
	Reflection	11 (18%)	7 (35%)	Capatina et al. (2018), Chaudy and Connolly (2019), Daoudi et al. (2021), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Minović et al. (2015), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018)
	Recommendation	10 (16%)	6 (30%)	Cengiz et al. (2017), Huang et al. (2019), Labonte-LeMoyne et al. (2017), Minović et al. (2015), Rodríguez-Cerezo et al. (2014), Slimani et al. (2018)
	Adaptation	5 (8%)	5 (25%)	Daghestani et al. (2020), Forsyth, Graesser, and Millis (2020), Labonte-LeMoyne et al. (2017), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Taub and Azevedo (2018)
	Prediction	3 (5%)	3 (15%)	Forsyth, Graesser, and Millis (2020), Konstantakopoulos et al. (2019), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017)

*Code (Anticipated objectives identified from reviewed studies) – N of codes (the occurrence of the anticipated objectives in reviewed studies).

Table 8. Additional objectives of LA use in online GBL in higher education.

Category	Objectives	N of codes* (%)	N of Studies (%)	References
Additional objectives	Engagement	12 (28%)	8 (40%)	Capatina et al. (2018), Daghestani et al. (2020), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Kuo and Chuang (2016), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Taub and Azevedo (2018)
	Motivation and emotion	6 (14%)	6 (30%)	Daoudi et al. (2021), Gañán et al. (2017), Kuo and Chuang (2016), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Taub and Azevedo (2018)
	Learning achievement	6 (14%)	6 (30%)	Huang et al. (2019), Muñoz-Merino et al. (2017), Palomo-Duarte et al. (2019), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018)
	Performance	5 (12%)	5 (25%)	Capatina et al. (2018), Daghestani et al. (2020), Loh, Li, and Sheng (2016), Rodríguez-Cerezo et al. (2014), Slimani et al. (2018)
	Decision-making	4 (10%)	4 (20%)	Capatina et al. (2018), Daghestani et al. (2020), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Loh, Li, and Sheng (2016)
	Interaction	3 (7%)	3 (15%)	Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Huang et al. (2019), Muñoz-Merino et al. (2017)
	Satisfaction	2 (5%)	2 (10%)	Gañán et al. (2017), Muñoz-Merino et al. (2017)
	Stress detection	1 (2.5%)	1 (5%)	Daoudi et al. (2021)
	Problem-solving	1 (2.5%)	1 (5%)	Hernández-Lara, Perera-Lluna, and Serradell-López (2019)
	Language learning	1 (2.5%)	1 (5%)	Palomo-Duarte et al. (2019)
	Self-regulation	1 (2.5%)	1 (5%)	Taub and Azevedo (2018)

*Code (Additional objectives identified from reviewed studies) – N of codes (the occurrence of the additional objectives in reviewed studies).

(Knight, Buckingham Shum, and Littleton 2013; 2020). A review of the included publications reveals that in GBL studies, LA is used for a wide range of purposes. In general, these findings suggest that LA can serve as an effective tool to support different types of assessment in online GBL in higher education.

4.3.2. Monitoring

A monitoring objective was identified in 40% of the studies (N of studies = 8) and accounted for 19% of the codes (N of codes = 12). LA as a data-driven field of study is a tracing tool that can track activities in the system (Banihashem, Noroozi et al. 2022). One of the LA purposes is monitoring learners' activities and learning processes, which helps educators to identify challenges that may impede learners' learning and better support learners with making evidence-based decisions and timely interventions (Chatti et al. 2012; Cornide-Reyes et al. 2020). In some reviewed studies, LA is used for monitoring purposes such as tracking and analysing learners' learning progress during a game (Mino-*vić* et al., 2015), metacognitive monitoring (Taub and Azevedo 2018), and tracking learners' communications and interactions in the game (Daghestani et al. 2020; Huang et al. 2019).

4.3.3. Reflection

Reflection was mentioned in 35% of studies (N of studies = 7) and 18% of the codes (N = 11) accounted for that purpose. In LA studies, learners become aware of their learning process and progress through reflections provided by LA (Kovanović et al. 2018). Studies have shown that LA reflections are effective in fostering

learning and critical thinking (Kim, Jo, and Park 2016; Koh, Jonathan, and Tan 2019). In a few of our reviewed papers, LA is used for reflection purposes (e.g. Capatina et al. 2018; Chaudy and Connolly 2019; Hernández-Lara, Perera-Lluna, and Serradell-López 2019). In one study, LA is used in a simulation game to reflect on what learners learn during gameplay (Hernández-Lara, Perera-Lluna, and Serradell-López 2019). In another study, Daoudi et al. (2021) adapted LA methods in the game to raise awareness among players.

4.3.4. Recommendation

In our set of studies, recommendation purpose was seen in 30% of studies (N of studies = 6) and 16% of the codes (N of codes = 10) identified for the recommendation. LA can provide insightful information to learners on what they should do next (Chatti et al. 2012). Since LA recommendations are based on analysing different types of data such as learners' history data and learning activity data, LA recommendations can be found useful for fostering learners' self-directed learning (Chatti et al. 2012). A few of the GBL-reviewed studies found that LA is used to provide recommendations and feedback (Cengiz et al. 2017; Labonte-LeMoyné et al. 2017; Rodríguez-Cerezo et al. 2014; Slimani et al. 2018). In these studies, LA worked as a recommender system. For example, Slimani et al. (2018) used LA to provide corrective feedback in the game for performance improvement. Similarly, Rodríguez-Cerezo et al. (2014) adopted LA as a tool for providing immediate feedback on successful and incorrect reactions. These findings suggest that LA not only provides insightful information on what has been done and how it has been done in the

past but also takes steps forward and intelligently recommends what should be done next.

4.3.5. Adaptation

Adaptation was found in 25% of studies (N of studies = 5) and 8% of the codes (N = 5) accounted for that purpose. LA employs educational data mining techniques to customise learning for individual learners by considering their learning needs (Daghestani et al. 2020). LA assists learners with adaptive learning by providing real-time feedback and reflections on the learning process (Lim et al. 2021). Some of the reviewed studies found that LA is useful in making GBL within higher education contexts adaptive (e.g. Daghestani et al. 2020; Forsyth, Graesser, and Millis 2020; Taub and Azevedo 2018). In one study, Daghestani et al. (2020) report that adaptive gamification supported by LA is effective in increasing learners' engagement and learning performance. In another study, Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017) report that adapting learning materials and game-based activities based on learners' preferences and interests can play a role in their time efficiency and learning behaviour. What can be understood here is that LA has the potential to support adaptive learning in GBL by employing artificial intelligence and educational data mining techniques.

4.3.6. Prediction

We found predication as an LA service for online GBL in 15% of studies (N of studies = 3) and 5% of the codes (N = 3) accounted for that purpose. LA serves as a tool to predict learners' learning outcomes based on their activities and behaviours (Greller and Drachslar 2012). According to the literature, LA methods and techniques are commonly used to predict learning behaviour and success and to give early warnings to learners at risk of failure (Arnold and Pistilli 2012; Dietz-Uhler and Hurn 2013). In a few of the reviewed GBL studies, LA is used for prediction purposes in online GBL (e.g. Forsyth, Graesser, and Millis 2020; Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos 2017). For instance, Forsyth, Graesser, and Millis (2020) adopted LA methods to predict learning in a multi-component serious game. Similarly, Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017) used LA methods for modelling learners' behaviour based on the earning badges in the game. In another study, Konstantakopoulos et al. (2019) found that LA methods can be used to forecast patterns of resource usage in a simulation game. These findings illuminate that LA not only describes what and how learners are

doing in the game but also can forecast the future based on the data generated during the gameplay.

As shown in Table 8, a review of the selected GBL studies reveals that there are other objectives for the use of LA in online GBL within higher education contexts that do not fit into LA objectives anticipated by Chatti et al. (2012). We call these findings additional objectives. Our review shows that LA is also used in online GBL in higher education for other purposes such as increasing engagement (N of studies = 8, 40%; N of codes = 12, 28%), increasing motivation, and discovering emotions (N of studies = 6, 30%; N of codes = 6, 14%), learning achievement (N of studies = 6, 30%; N of codes = 6, 14%), performance improvement (N of studies = 5, 25%; N of codes = 5, 12%), decision-making skills improvement (N of studies = 4, 20%; N of codes = 4, 10%), and discovering and improving the players' interaction (N of studies = 3, 15%; N of codes = 3, 7%). These additional objectives for LA use in GBL environments indicate that LA's application in higher education has grown beyond its expected reasons as anticipated by Chatti et al. (2012) and LA use affects education and learning in a wide range. We also note that these additional objectives can overlap with the anticipated objectives in Table 7 (e.g. Engagement in Table 8 might have involved Assessment, Monitoring, or Reflection in Table 7). This indicates that although we distinguished between anticipated and additional objectives based on our theoretical model to provide a clear overview of the LA objectives in online GBL studies, in practice these objectives are interrelated and difficult to separate from one another.

4.4. RQ4. Who are the stakeholders of using LA in online GBL studies in higher education?

In our set of GBL studies, we found 29 codes pointing to the stakeholders of LA used in online GBL within higher education contexts (Table 9). Unsurprisingly, learners were found to be the most common recipients of LA services in online GBL environments (N of codes = 19, 65%; N of papers = 19, 95%). Educators were also found to be served by LA in online GBL within higher education (N of codes = 8, 28%; N of papers = 8, 40%), and in a few reviewed studies, system designers were mentioned as another audience who benefited from LA use in online GBL in higher education (N of codes = 2, 7%; N of papers = 2, 10%).

A review of the included publications shows that LA serves learners, educators, and system designers for different reasons in the GBL environment within higher education contexts. For learners, LA serves as a reflection tool on the learning process and progress and lets

Table 9. Stakeholders of LA's application in online GBL in higher education.

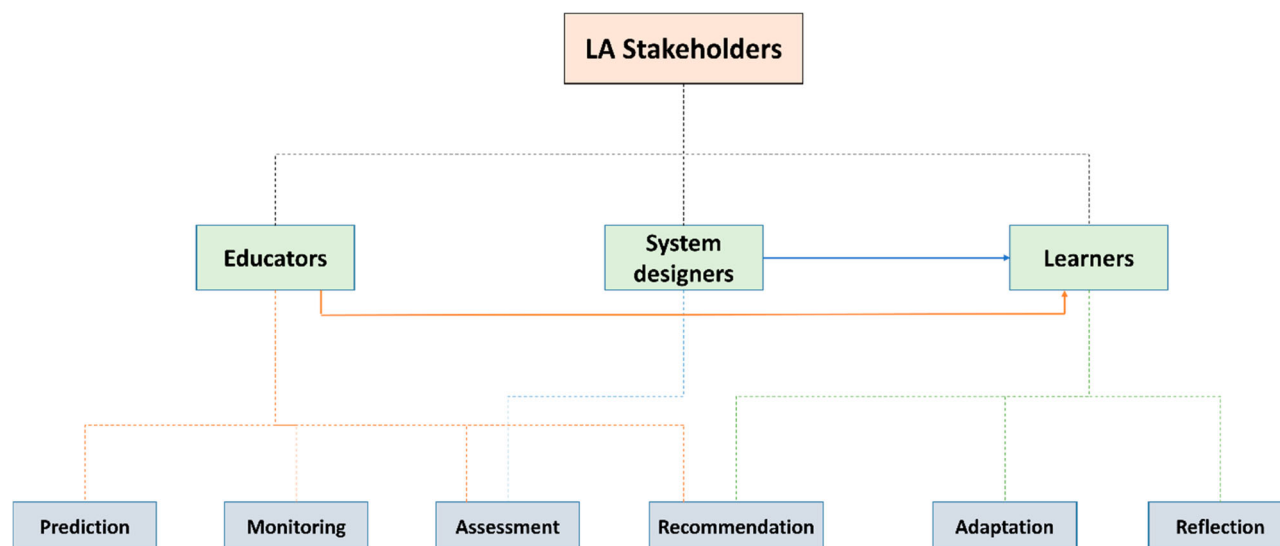
Stakeholders	N of codes (%)	N of Studies (%)	References
Learners	19 (65%)	19 (95%)	Capatina et al. (2018), Cengiz et al. (2017), Chaudy and Connolly (2019), Daghestani et al. (2020), Daoudi et al. (2021), Forsyth, Graesser, and Millis (2020), Gañán et al. (2017), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Huang et al. (2019), Kuo and Chuang (2016), Labonte-LeMoyne et al. (2017), Loh, Li, and Sheng (2016), Minović et al. (2015), Muñoz-Merino et al. (2017), Palomo-Duarte et al. (2019), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017), Slimani et al. (2018), Taub and Azevedo (2018)
Educators	8 (28%)	8 (40%)	Capatina et al. (2018), Chaudy and Connolly (2019), Gañán et al. (2017), Hernández-Lara, Perera-Lluna, and Serradell-López (2019), Kuo and Chuang (2016), Minović et al. (2015), Rodríguez-Cerezo et al. (2014), Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017)
System designers	2 (7%)	2 (10%)	Capatina et al. (2018), Chaudy and Connolly (2019)

*Code (Stakeholders identified from reviewed studies) – N of codes (the occurrence of the stakeholders in reviewed studies).

learners become aware of how they are doing (Daoudi et al. 2021; Hernández-Lara, Perera-Lluna, and Serradell-López 2019; Slimani et al., 2018). LA employs analytics methods to provide a dynamic and adaptive GBL environment where learners can follow and adjust their learning based on their own specific learning needs (Daghestani et al. 2020; Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos 2017). In some studies, it was found that LA serves as a recommendation tool for learners to know what to do next based on their activities and performance (Minović et al., 2015; Rodríguez-Cerezo et al. 2014). For educators, LA provides assessment services in which educators use LA outputs to assess learners' performance and learning in GBL (Chaudy and Connolly 2019; Gañán et al. 2017). In addition, LA serves as a monitoring tool for educators to track learners' interactions, behaviours, and performance in GBL environments (Capatina et al. 2018; Hernández-Lara, Perera-Lluna, and Serradell-López 2019). Some other studies showed that LA helps educators by

providing real-time feedback and recommendations (Minović et al., 2015; Rodríguez-Cerezo et al. 2014). A few studies showed that LA provides assessment services for system designers (Capatina et al. 2018; Chaudy and Connolly 2019). For example, LA provides insightful information regarding the profitability, usability, and cost-effectiveness of the designed game for system designers (Capatina et al. 2018; Chaudy and Connolly 2019).

Although our findings show that LA was used for different reasons by different stakeholders, it seems that the ultimate goal of LA in online GBL studies is to support learning. In some studies, however, this support comes from educators and system designers. For example, as described above, LA provides assessment and monitoring services for educators, but these services are provided for educators to monitor learning activity, identify gaps in learning during the gameplay, and assess their learning quality to provide appropriate interventions for learning improvement. In support of

**Figure 2.** Categorisation of LA objectives in GBL based on the stakeholders.

the findings we discuss here, a visual map provides a better illustration of the relationship between LA stakeholders and LA objectives in GBL studies (Figure 2).

5. Discussion

Our literature review found that different types of data were used in LA of the reviewed GBL studies. The majority of the data collected in GBL captures information about learners and their processes, collected either during the gameplay (e.g. time spent on the game, emotional and psychological reactions, number of attempts in the game) or from the player himself/herself (demographic data, psychological data, academic background data). This finding regarding learner-generated data implies that the use of LA in online GBL studies is mainly learner-centered. This result is supported by the literature that points out the consideration of learners' perspectives in LA (Ferguson 2012; Gašević, Dawson, and Siemens 2015). In line with this finding, although we found that educators and system designers have also benefitted from the LA services in GBL studies, these services primarily aim to increase the quality of learning and learner performance in GBL practices. For example, assessment services of LA for system designers aim to design a GBL environment that is more suited to learners' learning characteristics (Capatina et al. 2018; Chaudy and Connolly 2019). This is in line with our finding that LA data in GBL studies are learner-centered and is also consistent with the literature and the definition of LA where it is highlighted that LA is about learners and the main purpose of LA is to understand and optimise learning and learning environments (Gašević, Dawson, and Siemens 2015; Siemens and Long 2011). In general, this finding suggests that LA projects in GBL studies are directly or indirectly focused on supporting learners' learning and performance.

Our review reveals that different LA methods at different levels were used. Overall, data mining methods were the most commonly employed LA methods in GBL studies. This finding is supported by prior studies that point out the key role of data mining in LA systems (Chatti et al. 2012; Siemens and Baker 2012). The dominance of data mining methods in GBL studies, however, does not mean that these methods are sufficient on their own for running LA projects in GBL environments. That is to say that, in most LA projects in GBL studies, a multiple-method approach in mixed LA levels is employed. For example, data mining methods were used in the diagnostic and predictive analytics levels to cluster learners based on their performance and behaviour in the game, and visualisation methods

were used at a descriptive level to describe data mining results (Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos 2017). Another example is the combination of text analytics, data mining, and visualisation methods at the diagnostic level to explore learners' interaction patterns and assess learning (Hernández-Lara, Perera-Lluna, and Serradell-López 2019). This finding suggests that the optimal use of LA methods involves a mixed approach to LA methods and levels. The key point in terms of defining what kinds of methods at which level should be combined and used mainly depends on the nature of the data and the question that seeks to be answered by LA (Chatti et al. 2012).

We found that GBL studies used LA for different reasons, which we categorised into anticipated and additional LA objectives. This categorisation helped us to make a distinction between objectives drawn from our LA theoretical framework (Chatti et al. 2012) and objectives that were not fitted to the framework. These additional objectives indicate that LA is evolving; its potential to contribute to learning in GBL environments can grow beyond the LA scholars' prior anticipation. According to our set of studies, an example that illustrates the growing nature of LA is that most of these additional objectives were studied since 2018, such as engagement (Daghestani et al. 2020), motivation (Taub and Azevedo 2018), and stress detection (Daoudi et al. 2021). In addition, although in theory, we coded the LA services into anticipated and additional objectives, in practice they are normally interrelated. For example, in one study, LA contributed to learning achievement by providing recommendation and assessment services alongside identifying learners' patterns of interaction and engagement (Huang et al. 2019). This suggests that we should view LA use in GBL studies in a broader and combined way in terms of achieving objectives.

Regarding the context of higher education in this study, our findings reveal similarities and variations in comparison to previous review studies that encompassed settings beyond higher education such as primary and secondary education (e.g. Alonso-Fernandez et al. 2019a; Chaudy, Connolly, and Hainey 2014; Daoudi 2022). The similarities were found mainly in terms of the methods (how) between the current study and previous studies. For example, we reported data mining, visualisations, text mining, and social network analysis as the main LA methods used in LA-supported GBL in higher education, and almost similar methods are reported by previous studies (see Alonso-Fernandez et al. 2019a; Chaudy, Connolly, and Hainey 2014). This means that regardless of the context of education whether it is higher education, primary/secondary

education, or informal education, LA adopts certain methods and techniques. In addition, similarities were found regarding stakeholders (who) where educators, learners, and designers were reported as the main stakeholders (e.g. Alonso-Fernandez et al. 2019a; Chaudy, Connolly, and Hainey 2014). This means that the mission of LA is to support educators and learners and it does not change considering the given educational context. Furthermore, similarities can be found in terms of data (what) as our findings align with previous studies (e.g. Chaudy, Connolly, and Hainey 2014). For example, similar to our studies, Chaudy, Connolly, and Hainey (2014) also found data such as game data, player data, gameplay data, and time-related data as common data used by LA to support GBL. This can indicate that LA may use the same type of data to support GBL in different educational contexts. The main variation between the current study and previous studies was found regarding the purpose (why) of the use of LA. It seems that the purpose of using LA for GBL in higher education is broader and goes beyond expected LA objectives (e.g. recommendation, monitoring, reflection, assessment) and it includes varied objectives from language learning, motivation, and decision-making to higher-order skills such as self-regulation and problem-solving. While in previous studies, it was found that LA's purpose is more about assessment, providing timely interventions, or profiling learners (see Alonso-Fernandez et al. 2019a). Therefore, we can conclude that what makes LA to be different for various educational contexts is the reason why LA is used for. Our findings and discussion here are in line with and supported by our earlier argument in the introduction section where we pointed out that LA in higher education is more focused on higher-order skills compared to other educational settings (e.g. primary/secondary education).

5.1. Educational insights

The use of LA in GBL contexts could lead to technological determinism if implemented without considering the underlying theoretical foundations (Banihashem and Macfadyen 2021; Gašević, Dawson, and Siemens 2015; Knight, Shum, and Littleton 2014). This means that the application of LA in GBL contexts should be based on the pedagogical assumptions driven by theoretical foundations (Koh et al. 2016; Stewart 2017; Wise et al. 2016). Theory guides LA use in educational settings in three ways: (1) influencing the design and selection of LA tools, (2) guiding LA use by providing assumptions about learning processes, and (3) explaining learning behaviours and experiences, which help

interpret LA data (Banihashem et al. 2019). Below we explain how GBL is situated in various theories and discuss how our findings differ depending on the underlying theory.

Wu et al. (2012) highlighted the importance of using learning theories to guide GBL as an instructional strategy. They discussed the application of behaviourism, cognitivism, and constructivism in GBL research and practice.

They suggest that in GBL from a behaviourist perspective, players need to exactly know the game rules (e.g. what can be done and what cannot be done), and gameplay (e.g. what are the goals). From this perspective, players are treated as passively absorbing the narratives and being filled with concrete information (Wu et al. 2012). Based on these assumptions, LA should focus on reporting quantitative data from observable behaviours (Knight, Buckingham Shum, and Littleton 2013). LA tools should collect, analyse, and report data related to players' behaviour, such as gameplay data (e.g. time spent on the game, number of attempts, time spent on interaction, number of taken badges) and assessment data (e.g. grades, tests, fail/success) since they are directly related to player's behaviour. Visualisation and data mining methods are expected to be used more than text mining and social network analysis in the behaviourist-situated GBL context (Wu et al. 2012). The LA goals in this context should focus on decision-making, assessment, learning achievement, and performance, as behaviourism emphasizes learning outcomes over learning processes (Boghossian 2006).

Wu et al. (2012) explain that from a cognitivist perspective, learning in GBL involves cognitive thinking and game rules can range from simple to complex. The gameplay in such a GBL context focuses on the context-dependent nature of knowledge where players can control and build their learning at their own pace. The game narrative can be sophisticated and involve cognitive and emotional aspects. Based on these assumptions, LA tools tend to collect, analyse, and report more cognitive and emotional data, such as cognitive reactions, emotional reactions, and facial expressions of emotions. Player data (e.g. player characteristics, age, psychological features) is also important in cognitivist GBL contexts, as LA can use this type of data to provide insights into the reasons behind players' cognitive and emotional reactions. Data mining methods are the main LA methods in such contexts since they can explore cognitive developments and patterns and provide meaningful reports out of cognitive and emotional data. The goal of LA in cognitivist-situated GBL is focused on understanding how learning is happening and how a learner is doing, such as reflections, feedback,

monitoring, and discovering engagement patterns, motivation, and emotion.

Furthermore, Wu et al. (2012) suggest that in a constructivist-situated GBL, the focus is on the interaction between players and the game, and learning is a social process where the learner is at the centre. The game rules should be designed to facilitate player interaction, and the game narrative should be interactive and socially constructed. Based on these assumptions, LA should focus on capturing and analysing data related to the player's social learning process, such as interaction with the platform and other characters, discussion forum chat, and posts (Knight, Shum, and Littleton 2014). Social network analysis and text mining methods may be used to explore interaction patterns and forum posts. The goal of LA in a constructivist GBL context should focus on engagement, interaction, self-regulation, adaptation, and reflection, as constructivism views learning as a socially constructed process (Banihashem and Macfadyen 2021; Banihashem, Farrokhnia et al. 2022; Noroozi et al. 2019).

5.2. A conceptual framework for LA use in online GBL environments in higher education

The previous section discussed our results of the four main components of LA systems in GBL practices. Using these findings, we developed an evidence-based conceptual framework to assist in the use of LA in online GBL settings within higher education settings (Figure 3). To create this conceptual framework, we

reviewed previous LA models to see how the elements of LA systems should be arranged (see Chatti et al. 2012; Clow 2012; Elias 2011; Greller and Drachsler 2012). We then incorporated our own research findings into this framework in accordance with the suggestions made by these scholars. Essentially, we used a layering approach to build upon the existing knowledge and develop a more comprehensive framework for guiding the use of LA in online GBL environments within higher education contexts.

The proposed framework for the use of LA in online GBL studies highlights the importance of identifying the intended beneficiaries and objectives for the use of LA services. According to this framework, it is crucial to first determine who the target audience is, whether it is learners, educators, or system designers, as this information determines the required data for collection and the necessary methods for analytics. The objective of the LA services plays a crucial role in the rest of the process, as it defines the level of analytics and required LA methods (Banihashem, Noroozi et al. 2022; Chatti et al. 2012; Greller and Drachsler 2012). If the purpose of LA is not clearly defined, the entire LA process may be ineffective. Therefore, it is essential to establish the intended beneficiaries and objectives before implementing LA services in online GBL practices, as recommended by prior studies (e.g. Greller and Drachsler 2012).

The next step in the LA process after determining the intended beneficiaries and objectives is to identify the types of data that are required to meet those objectives.

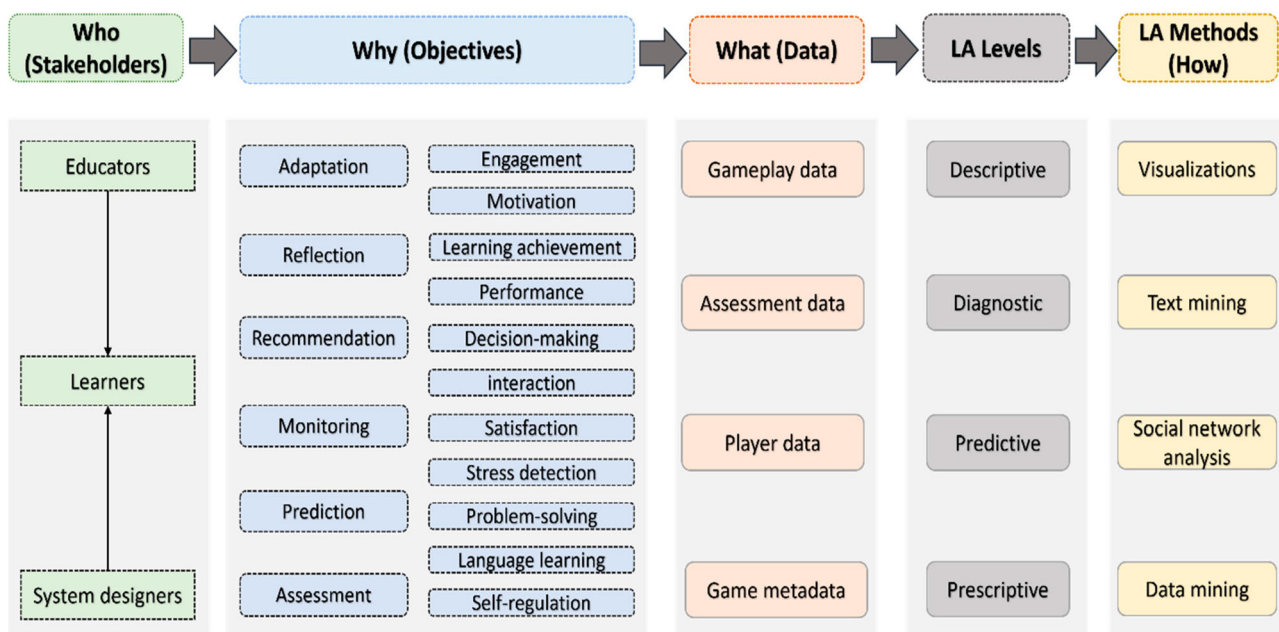


Figure 3. The conceptual framework for LA use in online GBL in higher education based on Chatti et al.'s model (2012).

Data is essential to the LA process as analytics rely on data (Siemens and Long 2011). Different types of data are available for different stakeholders and for different purposes. After determining the required data, the appropriate level and methods of analytics must be determined (Banihashem, Noroozi et al. 2022; Chatti et al. 2012). Different methods such as data mining, social network analysis, and visualisation methods can be employed at different levels of analytics depending on the purpose of LA (Eriksson, Bigi, and Bonera 2020; Henry and Venkatraman 2015; Maoz, 2013). For example, if the purpose of LA is to provide information on the current state of the player in the game, descriptive analytics with visualisation methods is appropriate. On the other hand, if the purpose is to identify the best performer in GBL, diagnostic analytics with visualisation and social network analysis methods are more suitable.

What can be concluded here is that to effectively apply LA, it is necessary to have a clear understanding of the intended beneficiaries and the objectives of the LA process. It is also essential to identify the types of data required to meet those objectives. The next step is to determine the appropriate level of analytics and methods to analyse the collected data. The appropriate application of LA requires a holistic understanding of all these factors. Without a clear understanding of these elements, the LA process may not produce meaningful insights or may even be counterproductive. Therefore, to achieve the desired outcomes from LA in GBL practices, it is crucial to have a systematic and evidence-based approach that considers all these factors.

5.3. Limitations and future research

There are a few limitations to this study that should be noted here. First, in this study, we did not include non-empirical studies such as review papers, conceptual papers, and theoretical papers, and we limited our review to only peer-reviewed papers and excluded other types of documents such as book chapters, dissertations, and lecture notes. In addition, we acknowledge our limitation for databases, however, we feel that our selected databases cover most of the relevant publications. Second, although we used different keywords for a search query to make sure that all relevant publications are covered, we acknowledge the likelihood that some papers may have not been found according to this search query. We thus acknowledge that our exclusion criteria might have resulted in removing some of the relevant publications. Future studies thus could expand the covered databases and search strategies to include non-empirical papers for a more

comprehensive analysis of LA in GBL environments. Third, although we provided an overview of game characteristics in selected studies and the role of LA in the games (see Appendix 1), this review did not extensively outline the specific utilisation of game elements in game-based learning (GBL) or elaborate on how these elements are interconnected with elements of learning analytics (LA). Essentially, our discoveries offer a general overview of the involvement of LA in GBL approaches. It would be intriguing for subsequent studies to thoroughly investigate the relationship between game elements and LA elements. Fourth, this study did not report any information on the barriers to and challenges of LA implementation in GBL environments within higher education contexts. We recommend this topic to be explored in future studies. Finally, in this study, we propose a conceptual framework to guide LA use in online GBL in higher education. Although the proposed framework was built on a well-known LA model and relevant prior empirical findings, it is not a prescriptive model to suggest explicit recommendations on how to guide LA use in GBL studies. It provides only more conceptual and general recommendations. Future studies should focus on designing a prescriptive model underpinning such a conceptual framework.

6. Conclusion

In this study, we systematically reviewed the literature on LA's role in online GBL within higher education contexts. Our results were built on the LA Reference Model suggested by Chatti et al. (2012) and we found that the four main elements of LA systems including what (types of data), how (levels and methods), why (objectives), and who (stakeholders) should be taken into consideration for effective delivery of LA in GBL environments. This review study provides insights into the core elements of the LA system and how these elements are connected with online GBL practices within higher education contexts to maximise learners' engagement and eventually learning. We have presented a comprehensive but general overview of the current state of LA's use in the GBL studies. More in-depth studies are required to examine the role of each element of LA in GBL practices. We have also developed a conceptual framework to guide LA's effective implementation for the enhancement of GBL practices in higher education. In the end, we conclude that the findings of this study contribute to extending our knowledge regarding the relations between LA and GBL fields of study and provide recommendations on how to use LA in GBL.

Acknowledgment

We would like to express our gratitude to Seyedahmad Rahimi from the University of Florida and Oleksandra Poquet from the Technical University of Munich for their invaluable feedback and editing assistance on this paper.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix 1. Game characteristics in selected papers.

Authors	Game's name	Game's elements	Learner's role	Game's goal	Content	Game's genre	LA role
Capatina et al. (2018)	Simbound	Competition, engagement, score, round	Exploratory discussion on how to manage active learning progress	Team rankings and learning progress	Marketing	Simulation	Knowledge gain through game-based analytics
Chaudy and Connolly (2019)	EngAGe	Badge, reward, feedback, quests	To reflect on the usefulness of the tool for designing and performing a game-based assessment. Answering some questions	Assessment, visualizing learning	Math	Roleplay	Monitoring, reflection, debriefing, and modification of the game.
Daghestani et al. (2020)	AGLS	Challenges and leaderboard, levels and badges, forum posts and chatting, navigation, external reading, and videos	Forum participating, chatting, levels completing, taking badges, completing challenges, viewing leaderboard, navigating, external reading, viewing videos	Increasing learners' engagement and learning performance	Data	Roleplay	Classification, adaptation, and prediction
Daoudi et al. (2021)	iScen Platform	Challenge, competence, profile	Participants (including player characters and non-player characters) have different roles namely coordinator, security responsible, warden, and deputy who should collaborate and coordinate actions to manage an emergency evacuation procedure	Improving learners' assessment and crisis management	Economic	Simulation	Visualization, emotional recognition
Forsyth, Graesser, and Millis (2020)	Operation ARA	Competition, score points, narration, personalization, videos, e-text, interacting with characters	Log into the game and completing questionnaire and interacting with virtual characters and receiving feedback from them	Improving learning	Psychology	Roleplay	Predicting deep and shallow learning
Gañán et al. (2017)	ICT-FLAG	Rewards, points, badges, leaderboard	Doing assignments and exercises	Provide assessment tools for investigating different actions among learners, teachers	N/A	Roleplay	Assessment
Hernández-Lara, Perera-Lluna, and Serradell-López (2019)	Cesim Global Challenge	Badge, competition	Contribute, share, and provide feedback by using online discussion forum	Fostering learning process	Business	Simulation	Monitoring, reflection, and assessment
Huang et al. (2019)	Moodle	Badges, points, leaderboard, feedback, challenge, collaboration	Participating in online discussion forum, completing assignments, quiz, commenting on others tasks	Improving learners' online discussion quantity and the depth discussion	Management	N/A	Predicting interaction patterns, monitoring learning process
Konstantakopoulos et al. (2019)	smartNTU	Point, reward	Doing assignments and sharing their idea and playing the game	Improving human-building interaction and energy efficiency in smart infrastructure	Energy	Social	Visualization, reflection, assessment
Kuo and Chuang (2016)	LOPUPA	points, leaderboard, challenges, rewards	Completing questionnaire, tasks, and assignments	Motivating engagement for online academic dissemination	Academic information	Roleplay	Analysis of behavior, reflection, monitoring
Labonte-LeMoyne et al. (2017)	ERPsim	Challenge	Sharing, commenting, completing questionnaire	Preparation for business expectations	Business	N/A	Visualization, reflection, monitoring
Loh, Li, and Sheng (2016)	GAD profiles	Point, reward, challenge	Challenge the maze as many time as the player choose, doing assignments	Problem solving	Decision-making	Puzzle	Visualization, reflection, monitoring
Minovic et al. (2015)	N/A	Quest, level, quiz	Solving all quests	Tracking learners' learning and improving learning process	Computer	Adventure	Visualization, reflection, assessment

(Continued)

Appendix 1. Continued.

Authors	Game's name	Game's elements	Learner's role	Game's goal	Content	Game's genre	LA role
Munoz-merino et al. (2017)	Khan Academy platform	Point, badge, collaboration	Watching videos, registering and logging in the platform, doing exercises	Improving learning	Physic	Roleplay	Visualization, reflection, assessment
Palomo-Duarte et al. (2019)	The Hidden Room	Avatar, competition, level	Completing individual learning tasks, cooperation in form of text chat	Improving learning	Language	Simulation	Visualization, classification, prediction, reflection
Rodríguez-Cerezo et al. (2014)	Evaluators	Level, avatar, feedback	Completing levels and assignments	Learning the fundamentals of semantic evaluation process behind attribute grammars	Computer	Simulation	Visualization, feedback, reflection
Ruipérez-Valiente, Muñoz-Merino, and Delgado Kloos (2017)	Khan Academy platform	Setting up goals, badges	Logging to the platform and following the steps of the game as designed especially interacting with badges	Identifying metrics that provide information regarding the behavior of learners with badges, including if they are intentionally earning them, the concentration for achieving them and their time efficiency.	Engineering	Roleplay	Visualization, clustering, interaction pattern identification, adaptation, reflection,
Slimani et al. (2018)	ELISA	Point, score	Interacting with the system, doing exercises	Improving performance	Biology	Simulation	Visualization, clustering, prediction, debriefing, reflection
Taub and Azevedo (2018)	CRYSTAL ISLAND,	Character, collecting different items, points, reward	Reading books and research papers, talking to non-player characters, and collecting food items to be tested for the disease's transmission source	Metacognitive monitoring and scientific inquiry based on levels of efficiency and emotions	Microbiology	Simulation	Visualization, emotion analysis, analysis of metacognition, monitoring
Cengiz et al. (2017)	Mine4Escape	Feedback, reward, rules, mystery, choice, challenge, point	Completing surveys and doing tasks	Learning data mining techniques in a fun and engaging way	Engineering	Adventure	Visualization, clustering, prediction, reflection